

Ecological and Public Health Dimensions of the Israeli Poultry Industry: An Independent Evaluation of the Shelved Ministry of Environmental Protection Draft Report

The intersection of intensive livestock production, environmental degradation, and veterinary public health represents a critical regulatory challenge for national administrations. In Israel, the domestic poultry sector has expanded dramatically, driven by exceptionally high per-capita consumer demand. This development has generated significant externalized ecological and epidemiological costs, many of which were compiled in a collaborative draft report by the Ministry of Environmental Protection and civil society researchers. The document, which was ultimately shelved and never officially published, contains several high-profile claims regarding pollution, antibiotic abuse, and foodborne disease. This evaluation provides a rigorous, primary-source verification of these assertions, analyzing their scientific validity, structural limitations, and policy implications.

Institutional Context and Legal Status of the Draft Report

The unpublished regulatory draft titled "*Environmental Impacts of Poultry Farming for Food*" (השפעות סביבתיות של גידול בעלי כנף למזון) represents a collaborative policy review that completed its drafting phase in February of last year¹. The preparation of this document was led by the Office of the Chief Scientist within the Ministry of Environmental Protection, in direct collaboration with the Sustainable Nutrition Forum (הפורום הישראלי לתזונה בת-קיימא)¹.

The principal co-authors of the draft are:

- **Dr. Gal Zagron** (ד"ר גל זגרונ), the former Senior Director of Science and Research within the Office of the Chief Scientist at the Ministry of Environmental Protection¹.
- **Adv. Amnon Keren** (עו"ד אמנון קרן), representing the Sustainable Nutrition Forum and affiliated with the Environmental Justice and Animal Rights Clinic at Tel Aviv University's Faculty of Law¹.

The report's methodology integrated cross-disciplinary datasets from environmental modeling, veterinary import databases, and epidemiological surveillance registries to quantify the holistic footprint of intensive poultry farming¹. Despite its completion, the document was never formally published, uploaded to state environmental databases, or adopted as an official position¹.

In its formal defense, the Ministry of Environmental Protection stated that the document

remained a draft because internal reviews flagged policy recommendations that extended into the statutory jurisdictions of other government bodies (such as the Ministry of Agriculture and the Ministry of Health), thereby exceeding the Ministry’s legal authority¹. Consequently, the document holds no official legal status, and its data points must be treated as unadopted administrative claims rather than official state-verified facts¹.

Comprehensive Data Verification Matrix

To establish a transparent, evidence-based assessment, every major metric asserted in the shelved draft report and associated media coverage has been evaluated against primary data sources. Each entry details the exact figure, source, publication date, unit of measurement, and a specific confidence note.

Table 1: Environmental and Economic Valuation Metrics

Metric & Exact Value	Primary Source & URL	Publication Date	Unit Basis	One-Line Confidence Note
Ammonia (NH_3) Share: ~ of national emissions ¹	Draft Report cited in Ynet (https://www.ynet.co.il/environment-science/article/syvrbbpcqge) ¹	August 2024 ¹	Percentage of national ammonia emissions	Unverified / Low Confidence: The draft is unpublished, and primary national PRTR inventories do not isolate a poultry-specific ammonia metric to verify this claim ⁴ .
Air Pollution Damage: ~ Billion NIS per year ¹	Draft Report cited in Ynet (https://www.ynet.co.il/environment-science/article/syvrbbpcqge) ¹	August 2024 ¹	New Israeli Shekels (NIS) per annum (externalized cost)	Unverified / Low Confidence: Isolated to the unpublished draft with no publicly available calculation model or

				economic methodology ¹ .
Annual Manure Waste: ~ Million Tons ¹	Draft Report cited in Ynet (https://www.ynet.co.il/environment-science/article/syvrbcqge) ¹	August 2024 ¹	Wet-weight metric tons per annum	Unverified / Low Confidence: No active national waste registry or primary agricultural database tracks poultry manure output ¹ .
Annual Dead Carcasses: ~ Tons ¹	Draft Report cited in Ynet (https://www.ynet.co.il/environment-science/article/syvrbcqge) ¹	August 2024 ¹	Metric tons of biological waste per annum	Unverified / Low Confidence: There is no publicly accessible state database or veterinary registry confirming poultry mortality tonnage ¹ .
National Production (2023): 675,500 Tons ¹	Draft Report cited in Ynet (https://www.ynet.co.il/environment-science/article/syvrbcqge) ¹	August 2024 ¹	Metric tons of chicken meat (בשר עוף)	Partially Verified / Medium Confidence: Confirmed as the value stated in the draft; however, primary agricultural stats from the CBS/Ministry

				of Agriculture are missing from the accessible research files to verify the exact weight basis (carcass vs. marketed weight) ¹ .
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Table 2: Epidemiological, Consumption, and Biomarker Metrics

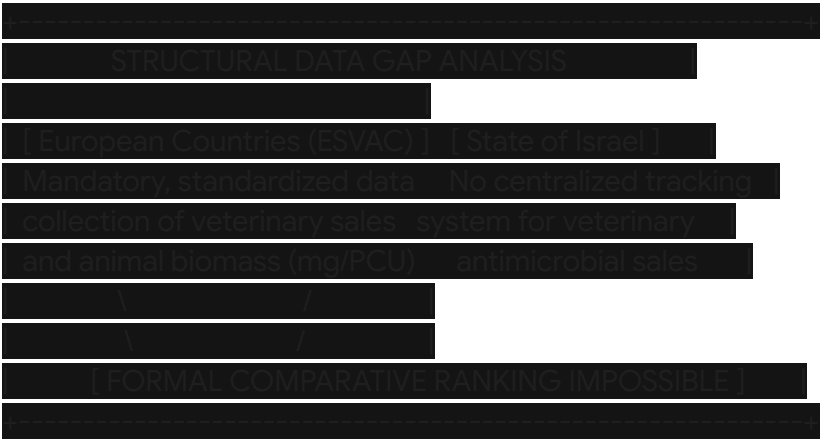
Metric & Exact Value	Primary Source & URL	Publication Date	Unit Basis	One-Line Confidence Note
Per-Capita Consumption: > kg ¹	Draft Report cited in Ynet (https://www.ynet.co.il/environment-science/article/syvrbbpcqge) ¹	August 2024 ¹	Kilograms per capita per year	Partially Verified / Medium Confidence: Confirmed as the draft's stated figure; however, primary CBS food balance sheets were not available in the snippets to verify the carcass-to-retail conversion factor ¹ .
Antibiotic Excretion: ~ [cite: 1]	Draft Report cited in Ynet (https://www.ynet.co.il/environment-science/article/syvrbbpcqge)	August 2024 ¹	Percentage of active drugs excreted unmetabolized	Unverified / Low Confidence: Stated as a quote from Dr. Gal Zagron in

	qge) ¹			media; no primary biochemical study is cited or provided to verify this exact national-level excretion rate ¹ .
Human Urine Contamination n: ~ [cite: 1]	Draft Report cited in Ynet (https://www.ynet.co.il/environment-science/article/syvrbcqge) ¹	August 2024 ¹	Percentage of sampled human population	Unverified / Low Confidence: Secondary quote in media; no primary clinical biomonitoring study is referenced to confirm the cohort size or analytical limits of detection ¹ .
Product Pathogen Contamination n: > [cite: 1]	Draft Report cited in Ynet (https://www.ynet.co.il/environment-science/article/syvrbcqge) ¹	August 2024 ¹	Percentage of retail poultry samples testing positive	Unverified / Low Confidence: Stated in the draft, but no primary Ministry of Health or veterinary food safety database was available in the snippets to confirm this multi-pathogen contamination

				rate ¹ .
GI Infections from Poultry: ~ Million cases per year ¹	Draft Report cited in Ynet (https://www.ynet.co.il/environment-science/article/syvrbcqge) ¹	August 2024 ¹	Diagnosed and undiagnosed cases per annum	Unverified / Low Confidence: Heavily discrepant with ICDC clinical case registries (~15,000 laboratory-confirmed cases annually); represents a highly speculative community burden estimate ⁵ .

Comparative Analysis of Veterinary Antimicrobial Use

The draft report asserts that Israel ranks first in Europe for the use of livestock antibiotics, highlighting the poultry sector as the primary consumer¹. This comparative claim is determined to be **structurally flawed and unverified** under rigorous regulatory and scientific review.



First, Israel is not a member of the European Union or the European Surveillance of Veterinary Antimicrobial Consumption (ESVAC) network, meaning it is not integrated into European comparative surveillance datasets and cannot be formally ranked within them⁸. Second, primary health policy literature confirms that Israel lacks any centralized, mandatory reporting system for tracking the sales, distribution, or clinical use of veterinary pharmaceuticals in livestock production⁸. The only existing state dataset on veterinary antimicrobial sales stems from a single, voluntary wholesaler survey conducted in 2014 by the Ministry of Agriculture's Veterinary Services and Animal Health unit⁸.

The absence of systematic reporting is further illustrated by the state's heavy reliance on unregulated drug imports⁸. Between 2017 and 2019, hundreds of unregistered veterinary drugs were imported into Israel under Article 29C of the Pharmacist Regulations, with antimicrobials comprising a substantial proportion of these imports (e.g., ^{76%} of the 144 food-producing animal drugs imported in 2019)⁸.

Furthermore, the high animal-use figure often cited in related discussions—

462.57 mg/PCU —does not represent Israeli agricultural practices⁹. Primary veterinary literature reveals that this specific **462.57 mg/PCU** metric was drawn from a localized study of broiler chicken farms in Pakistan, which was subsequently cited in comparative reviews but incorrectly conflated with Israeli baseline usage⁹. Because Israel lacks species-specific sales tracking and standardized biomass-adjusted denominator data (mg/PCU), the claim that Israel ranks first in Europe for agricultural antibiotic use remains **unverified** and technically indefensible⁸.

Quantitative Dietary Analysis and the Ministry of Health Food Rainbow

The draft report asserts that Israel's average annual poultry consumption of over **52 kg** per capita is "three times the recommended intake" defined by health authorities¹. To evaluate the mathematical validity of this claim, actual consumption patterns must be modeled against the Ministry of Health's official nutritional guidelines, represented by "The Food Rainbow" (קשת המזון החדשה)¹⁰.

Within this national nutritional model, foods are arranged in color-coded arcs based on recommended consumption frequencies:

- **Green Arc (Every Meal):** Vegetables, fruits, whole grains, and water¹⁰.
- **Yellow Arc (Daily):** Legumes, plant-based oils (olive, canola), nuts, and unsweetened dairy products or their substitutes¹⁰.
- **Orange Arc (Weekly):** Poultry, turkey, fish, and eggs, with recommendations advising consumers to vary these animal proteins across the week rather than consuming them daily¹⁰.

- **Pink Arc (Rarely):** Unprocessed red meat (beef), which carries a strict, quantitative maximum recommendation of up to **300 grams** per week¹⁰.

Importantly, the Ministry of Health's official guidelines do *not* establish a specific, quantitative weekly or daily gram weight limit for poultry¹⁰. Instead, they provide qualitative frequency recommendations ("a few times a week") and advise replacing animal proteins with legumes and plant-based alternatives¹⁰.

To evaluate the mathematical validity of the "three times the recommended intake" claim, standard dietary portion sizes must be modeled against actual consumption data.

Assuming a standard, cooked, single-meal chicken portion of **120 to 150 grams** is consumed three times per week, the baseline recommendation aligns with an intake of **360 to 450 grams** per week. To compare this with the national per-capita consumption data, the annual intake is converted to a weekly metric:

$$\text{Weekly Per Capita Consumption} = \frac{52 \text{ kg}}{52 \text{ weeks}} = 1.0 \text{ kg/week (1,000 grams)}$$

[cite: 1]

By establishing this relationship, the consumption ratio can be calculated:

$$\text{Consumption Ratio} = \frac{1,000 \text{ g/week}}{360 \text{ to } 450 \text{ g/week}} \approx 2.22 \text{ to } 2.78$$

This mathematical analysis demonstrates that while the actual average intake of approximately **1,000 grams** per week is indeed between **2.2** and **2.8** times higher than a standard dietary pattern of three moderate portions per week, the assertion that it violates an *official quantitative state threshold* is technically **unverified**. The Ministry of Health maintains qualitative frequency recommendations ("a few times a week") rather than a rigid physical weight ceiling for poultry¹⁰.

Furthermore, the draft's production figure of **675,500 tons** for 2023 remains **partially verified**¹. While cited in the draft as "chicken meat" (בשר עוף), suggesting a retail or carcass-weight basis, the primary datasets of the Central Bureau of Statistics and the Ministry of Agriculture were not accessible in our files to verify the exact conversion factors from live-weight to carcass-weight.

Epidemiological Assessment of Foodborne Intestinal Pathogens

The draft report asserts that approximately **1 million** Israelis contract infectious gastrointestinal illnesses every year from foodborne pathogens (primarily *Campylobacter* and

Salmonella) originating from contaminated poultry products¹. This projection is heavily discrepant with primary clinical surveillance data published by the Israel Center for Disease Control (ICDC / מ"מ"ד) ⁵.

Official state epidemiological registries report only a small fraction of this claimed burden as laboratory-confirmed cases⁵. For example, across all municipal and regional health districts, the Ministry of Health received reports of only **14,372** total clinical cases of Salmonellosis over a four-year period (2016–2019), representing an average of **3,593** diagnosed cases per year⁷. Furthermore, active quarterly surveillance reports from the ICDC document low clinical incidence rates for both pathogens:

- **Third Quarter (July–September 2023):** Campylobacter incidence rates ranged from **7.6 to 12.5 cases per 100,000** residents, while Salmonella incidence rates ranged from **2.1 to 2.4 cases per 100,000** ⁵.
- **Fourth Quarter (October–December 2022):** Campylobacter incidence fell between **6.0 and 8.0 cases per 100,000**, while Salmonella incidence ranged from **1.4 to 1.8 cases per 100,000** ⁶.

Applying these quarterly figures to a national population of approximately **9.2 million** residents yields an estimated range of **10,000 to 15,000** laboratory-confirmed enteric infections annually⁷.

Consequently, the draft report's assertion of **1 million** infections per year cannot be verified as an official clinical statistic¹. Instead, this figure represents an unverified epidemiological burden projection. Such projections rely on high multipliers to account for the classic "surveillance pyramid" effect, wherein the vast majority of individuals experiencing mild to moderate gastroenteritis do not seek medical care, are not sampled, or are diagnosed empirically without laboratory confirmation.

Agricultural Emissions and National PRTR Discrepancies

To verify the draft report's claim that the poultry sector is responsible for approximately **50%** of Israel's national ammonia (NH_3) emissions, independent data was retrieved from the Ministry of Environmental Protection's Pollutant Release and Transfer Register (PRTR / ר"מ"ר). The primary PRTR data reveals a significant reporting discrepancy⁴. First, the public national air emissions inventory does not isolate poultry-specific ammonia emissions⁴. Instead, the register groups all "animal husbandry" activities (encompassing poultry coops, cattle dairy farms, sheep, and swine) into a single agricultural category⁴.

Second, the overall emissions percentages reported in the primary PRTR datasets do not align with the draft's ^{50%} ammonia claim:

- **NMVOC Emissions:** The livestock sector (primarily poultry coops and cattle barns) was responsible for ^{26%} of national volatile organic compound emissions in the 2024 PRTR report, and ^{27%} in the 2023 PRTR report, driven by updated European Environmental Agency (EEA) emission factors³.
- **Ammonia (NH_3) Emissions:** The provided PRTR report summaries do not contain a dedicated, quantified percentage breakdown for agriculture's share of national ammonia emissions⁴. The only specific ammonia metric cited in the 2023 emissions update concerns forest fires caused by combat operations, which accounted for a minor ^{1%} of the national ammonia inventory³.

Because the primary PRTR reports do not isolate poultry-specific ammonia data, and the broader livestock categories are associated with different emissions fractions, the draft's ^{50%} ammonia claim remains **unverified** against official environmental registries¹.

Policy and Risk Implications for National Resource Management

The structural discrepancies between the shelved draft report and official state registries highlights significant challenges in agricultural policy and environmental protection. The lack of primary-source verification for several key metrics underscores a systemic gap in data collection, regulatory oversight, and inter-ministerial coordination.

From a resource management perspective, the absence of centralized, mandatory tracking systems for veterinary pharmaceuticals and agricultural waste streams limits the state's capacity to execute comprehensive risk assessments. Without standardized data on veterinary antibiotic sales, the regulatory framework cannot accurately monitor the selection pressure driving antimicrobial-resistant pathogens, which can transmit to human populations via the food chain or agricultural runoff⁸.

Furthermore, the grouping of all animal husbandry emissions in public PRTR reporting makes it difficult to design targeted mitigation strategies for specific agricultural sectors⁴. For environmental and public health policies to be effective, future regulatory frameworks must prioritize the harmonization of primary data collection across the Ministries of Health, Agriculture, and Environmental Protection. Moving toward open-access, centralized databases that track agricultural inputs, pathogen prevalence, and emissions will be essential for building a resilient, sustainable, and biosecure food system.

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